02.10.2025

- Build and Fix
- waterfall
- Rapid prototyping

Testing phase

└ unit testing

└ Integration Testing

System Testing.

- Waterfall Model ⇆ linear Sequential model

↓

here drawback:

client is not considered - Intervention is important

↳ Rapid prototyping model

- here prototype is added.

before requirement.

↓

- requirements from client

- After requirement, taking feedback from the client

(confirmation)

03.10.2025 Friday

Incrementive/Itrative Model

- Made build by buid/component by component.
  - Fit well and perfect delivery

Architecht of the system is the key.
-ure

- pros:
  - early delivery
  - No repetetive errors
  - More than 90% issues are resolved.
- Indirect Advantage
  - All teams are involved in the process.
  - Requirements should be well-defined.

Requirement + Architecture + Specification

↳ one unit.

SPIRAL MODEL

Risk Analysis
(Always expensive)

- plan mitigation strategies to reduce the impact of anticipating issues/problems.

- project Risks
- Technical Risks
- Business Risks
- Operational Risks

RISK — Identification
- Assessment
- Prioritization

- Mitigation

  - Acceptance.

XP

Architechtural Spike +

User Story

(Requirement)

criteria tobe defined earlier (formulate test cases)

↓

Planning → Spike

- certain
- uncertain

↓

Iteration

↓

Acceptance test

(repeat till requirements be fulfilled)

once 1st build is done, will be moved to next iteration

(model be prepared for releases)

• project velocity → progress of the project

16.10.2025

AGILE

— open to change/adapt (mindset)

Agile umbrella

XP, Scrum, Crystal Methods, DSDM, Kanban, FDD, Scrumban

more than 50% - SCRUM is used

• where requirements are clear, agile may not be suitable.

Project life cycle:

- predictive — fixed
- iterative
- Incremental (Requirements Defined)
- Agile / change Driven

time boxing

requirements may change

-user story-

EPIC - large story that can be divided

THEME - collection of related stories

Stories — actual stories

eg;

Improve Student Experience *theme*

↓

Develop student portal *Epic*

↓

View Grades *story*

Operational	warehouse
- Entries are done	- data retrieval is done

Types of requirements:

- Functional → core Features
- Non-Functional → whole system
- Inverse → what system should not do
- Domain
- Design & imp constraints
  ↓
  implementation

→ Technology
→ Range

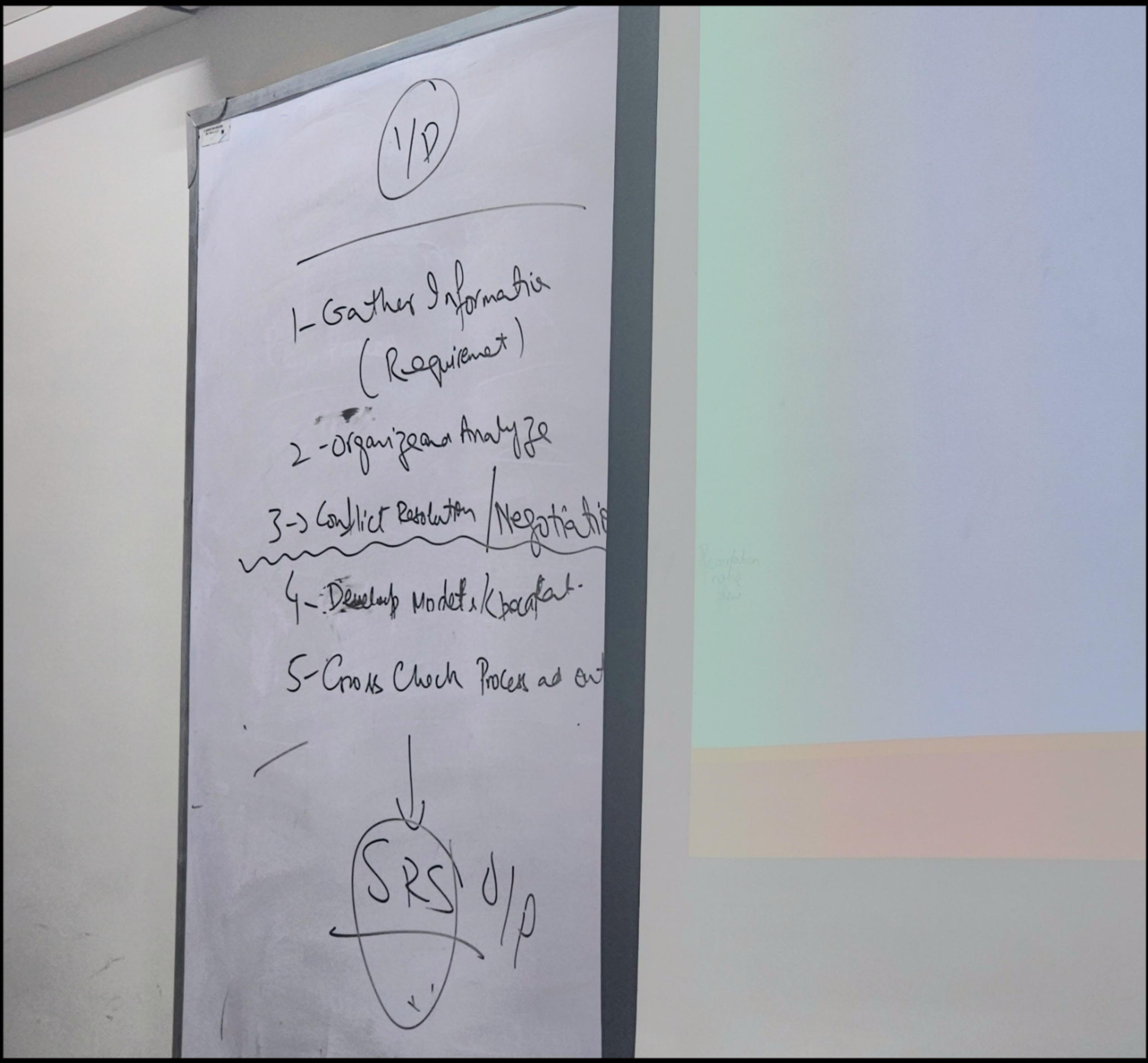
Requirement Engineering process

- Existing systems Information
- Stakeholders Need
- organizational standards
- Regulation
- Domain Information

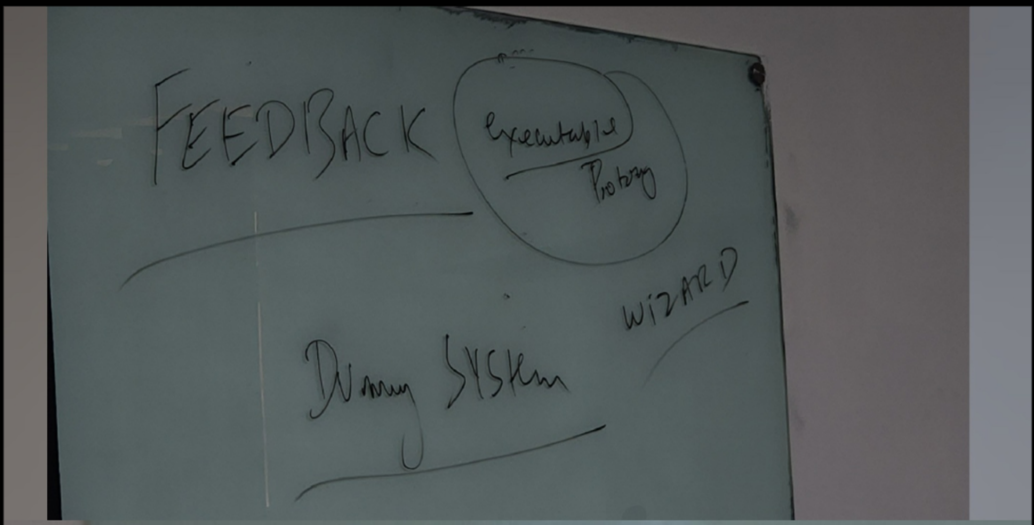
$\underbrace{\hspace{10em}}$

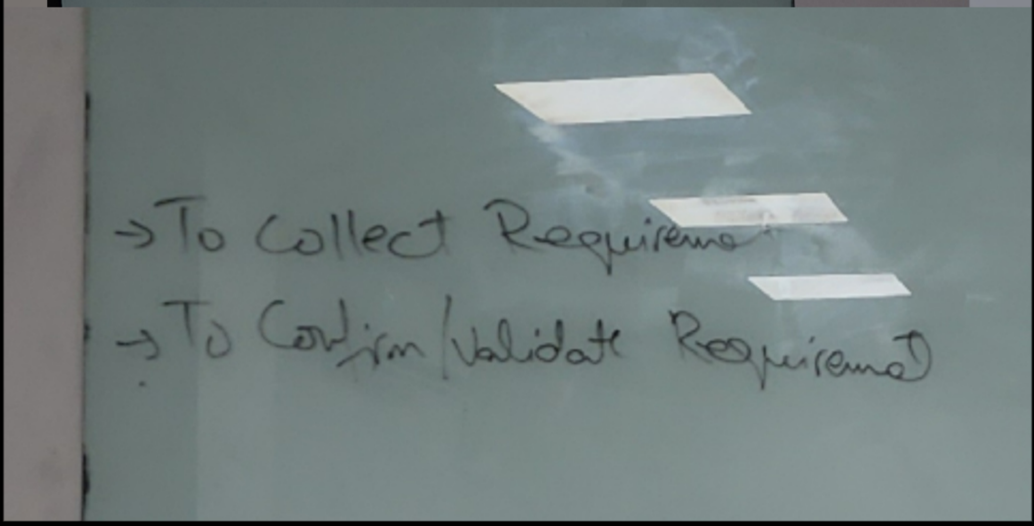
Requirement Engineering process

- Agreed Requirements
- System Specification
- System Models



30·10·2025





Types Of Prototype:

- Throw away prototyp
  - discard after feedback
  - unclear parts
- Evolutionary feedback
  - create initial version of system following all standereds
  - simple and most understandable parts
  - actual operational system.
    - ↳ operational prototype

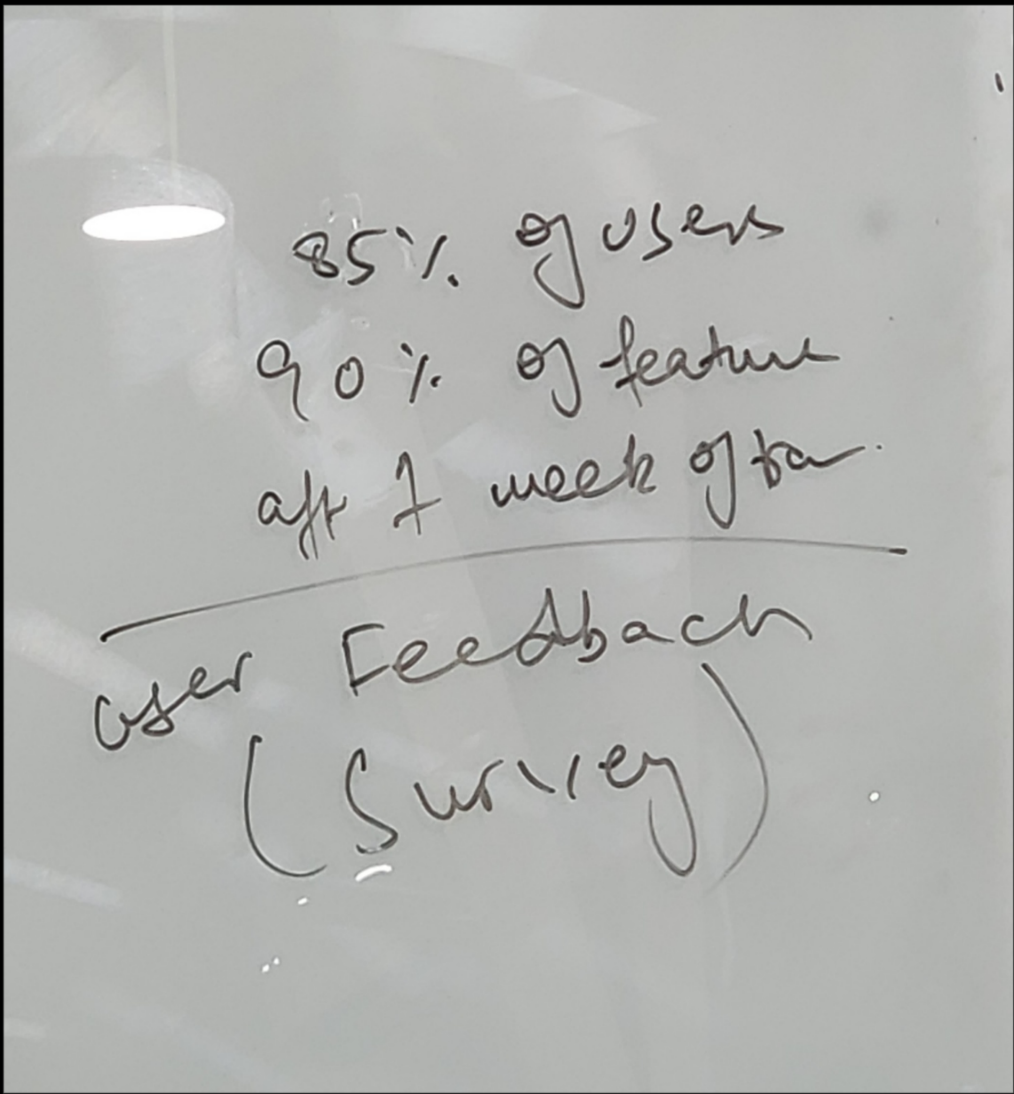
- CMM
  - SEI → Software engineering Institute
  - 5. ↳ Optimising (continuous improvement)
  - 4. → Managed (analysing the process)
  - 3. → Defined (all engineering and mgmt)
  - 2. → Repeatable (All processes related to Schedule and cost)
  - 1. ↳ Initial (Ad Hoc)

Engineering

Process
↳ Input to output following a set of steps.

Process Definition

↳ Writing it down



Types of Error:

- Commition
- Omission
- Clarity & Ambiguity
- Speed & capacity

Severity :

- Cosmetic Errors
- Minor Function Error
- Major Error
- Complete Failure / System Crash